

Chapter 10

Light: Reflection and Refraction

1. Light travels in a:

- A) Zig-zag path
- B) Straight line
- C) Circular path
- D) Random direction

Answer: B) Straight line

2. The reflection from a smooth, shiny surface like a mirror is called:

- A) Diffused reflection
- B) Regular reflection
- C) Irregular reflection
- D) Multiple reflection

Answer: B) Regular reflection

3. The angle of incidence is equal to:

- A) Angle of deviation
- B) Angle of reflection
- C) 90°
- D) 0°

Answer: B) Angle of reflection

4. The image formed by a plane mirror is always:

- A) Real and inverted
- B) Virtual and erect
- C) Real and erect
- D) Diminished and inverted

Answer: B) Virtual and erect

5. The focal length of a plane mirror is:

- A) Zero
- B) Infinite
- C) Equal to radius of curvature
- D) Equal to one

Answer: B) Infinite

6. The center of a spherical mirror is called:

- A) Focus
- B) Pole
- C) Center of curvature
- D) Aperture

Answer: C) Center of curvature

7. The distance between pole and focus of a mirror is called:

- A) Aperture
- B) Radius of curvature
- C) Focal length
- D) Depth

Answer: C) Focal length

8. For a concave mirror, the focal length is taken as:

- A) Positive
- B) Negative
- C) Zero
- D) Variable

Answer: B) Negative

9. For a convex mirror, the focal length is taken as:

- A) Positive
- B) Negative
- C) Zero
- D) Infinite

Answer: A) Positive

10. The mirror used by dentists to see enlarged images is:

- A) Plane mirror
- B) Concave mirror
- C) Convex mirror
- D) Parabolic mirror

Answer: B) Concave mirror

11. The mirror used in vehicles as rear-view mirror is:

- A) Plane mirror
- B) Concave mirror
- C) Convex mirror

D) None of these

Answer: C) Convex mirror

12. The mirror formula is:

A) $1/f = 1/v + 1/u$

B) $1/f = 1/u + 1/v$

C) $1/f = 1/u - 1/v$

D) $1/f = 1/v - 1/u$

Answer: D) $1/f = 1/v - 1/u$

13. The magnification formula is:

A) $m = v/u$

B) $m = u/v$

C) $m = h_1/h_2$

D) $m = f/u$

Answer: A) $m = v/u$

14. If the magnification is negative, the image is:

A) Virtual and erect

B) Real and inverted

C) Real and erect

D) Virtual and inverted

Answer: B) Real and inverted

15. If the magnification is positive, the image is:

A) Real and inverted

B) Virtual and erect

C) Real and erect

D) Diminished

Answer: B) Virtual and erect

16. The image formed by a convex mirror is always:

A) Real and inverted

B) Virtual and erect

C) Real and magnified

D) Inverted and enlarged

Answer: B) Virtual and erect

17. The image formed by a concave mirror when the object is placed between focus and pole is:

- A) Real and inverted
- B) Virtual and erect
- C) Real and diminished
- D) Real and magnified

Answer: B) Virtual and erect

18. Refraction occurs because:

- A) Light bends near opaque objects
- B) Speed of light changes in different media
- C) Wavelength of light changes
- D) Frequency of light changes

Answer: B) Speed of light changes in different media

19. The bending of light when it passes from one medium to another is called:

- A) Reflection
- B) Dispersion
- C) Refraction
- D) Diffraction

Answer: C) Refraction

20. The refractive index (n) of a medium is given by:

- A) $n = \sin i / \sin r$
- B) $n = \sin r / \sin i$
- C) $n = c/v$
- D) Both A and C

Answer: D) Both A and C

21. When a ray of light enters a denser medium, it:

- A) Bends towards the normal
- B) Bends away from the normal
- C) Goes straight
- D) Gets reflected

Answer: A) Bends towards the normal

22. When a ray of light enters a rarer medium, it:

- A) Bends towards the normal
- B) Bends away from the normal
- C) Gets absorbed

D) Gets reflected

Answer: B) Bends away from the normal

23. The phenomenon of refraction through a glass slab causes:

A) Lateral displacement

B) Dispersion

C) Internal reflection

D) Diffraction

Answer: A) Lateral displacement

24. A concave lens is also known as:

A) Converging lens

B) Diverging lens

C) Plane lens

D) Reflecting lens

Answer: B) Diverging lens

25. A convex lens is also known as:

A) Converging lens

B) Diverging lens

C) Reflecting lens

D) Plane lens

Answer: A) Converging lens

26. The lens formula is:

A) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

B) $\frac{1}{f} = \frac{1}{u} - \frac{1}{v}$

C) $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

D) $f = u \times v$

Answer: A) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

27. The magnification produced by a lens is:

A) $m = v/u$

B) $m = h_2/h_1$

C) Both A and B

D) None

Answer: C) Both A and B

28. A convex lens forms a virtual image when the object is placed:

- A) At focus
- B) Beyond 2F
- C) Between focus and optical center
- D) At infinity

Answer: C) Between focus and optical center

29. Power of a lens (P) is given by:

- A) $P = 1/f$ (in meters)
- B) $P = f^2$
- C) $P = 1/v$
- D) $P = 1/u$

Answer: A) $P = 1/f$ (in meters)

30. The unit of power of lens is:

- A) Meter
- B) Focal length
- C) Dioptre
- D) Centimeter

Answer: C) Dioptre